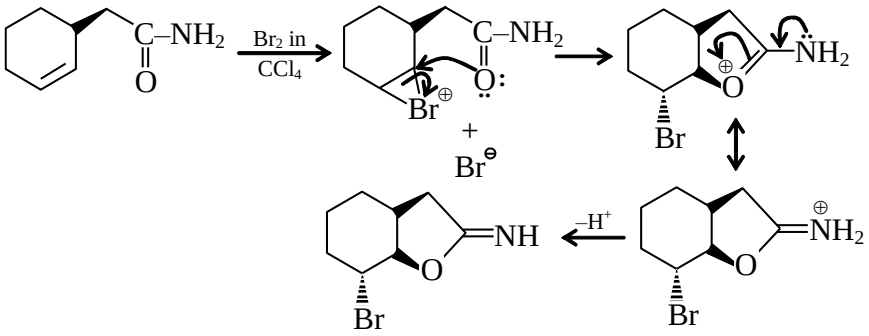
CHEMISTRY

SECTION – I

Single Correct Choice Type

1. (c)



2. (d)

$$HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$$

Initially	c	0	0
At equilib.	$c(1-\alpha)$	$c\alpha$	$c\alpha$

$$K_a = \frac{(c\alpha)^2}{c(1-\alpha)} = \frac{(0.1\,c)^2}{0.9\,c} = \frac{0.01\,c}{0.9}$$

$$K_a = \frac{0.1\,c}{9} \quad ; \quad c = \frac{9\,K_a}{0.1} = \mathbf{90\ K_a.}$$

3. (c)

$$A \underset{k_r}{\overset{k_f}{\rightleftharpoons}} B \quad \text{(Mutarotates)}$$

t = 0	a	0
t = t	a – *x*	*x*
t = ∞	a – y	y

Rate of forward reaction, $\frac{dx}{dt} = k_f(a - x)$

Rate of reverse reaction, $\frac{-dx}{dt} = k_r x$

Observed rate of reaction, $\frac{dx}{dt} = k_f(a - x) - k_r x$

$$= k_f a - (k_f + k_r)x \qquad \ldots(1)$$

Let, 'y' be the value of *x* at equilibrium.

At equilibrium, rate of forward reaction = rate of reverse reaction.

$$k_f(a - y) = k_r y$$

$$k_f a - k_f y = k_r y$$

$$a = \frac{(k_f + k_r)}{k_f}.y$$

Substituting value of 'a' in equation (1), gives

$$\frac{dx}{dt} = k_f \frac{(k_f + k_r)}{k_f} y - (k_f + k_r)x = (k_f + k_r)(y - x)$$

$$\int_0^x \frac{dx}{(y - x)} = (k_f + k_r)\int_0^t dt$$

$$\left[-\ln(y - x)\right]_0^x = (k_f + k_r)\, t$$

$$-\ln (y - x) + \ln y = (k_f + k_r)\, t$$

$$(k_f + k_r) = \frac{1}{t} \ln \frac{y}{(y - x)}$$

Let r_A and r_B be the optical rotations for 1 mole of A & B respectively.

$r_o = ar_A$

$r_t = (a - x)r_A + xr_B = ar_A + x(r_B - r_A) = r_o + x(r_B - r_A)$

$$x = \frac{(r_t - r_o)}{(r_B - r_A)}$$

$r_\infty = (a - y)r_A + yr_B = ar_A + y(r_B - r_A) = r_o + y(r_B - r_A)$

$$y = \frac{(r_\infty - r_o)}{(r_B - r_A)},\ (y - x) = \frac{(r_\infty - r_t)}{(r_B - r_A)}$$

$$\therefore \quad \frac{y}{y - x} = \frac{(r_\infty - r_o)}{(r_\infty - r_t)} = \frac{30 - 190}{30 - 150} = \frac{-160}{-120} = \frac{4}{3}.$$

$$\therefore \quad (k_f + k_r) = \frac{2.303}{180} \log \frac{4}{3} = \frac{2.303}{180} \times 0.123$$

$(k_f + k_r) = 1.57 \times 10^{-3}$ min^{-1}.

4. **(c)** Bonding in I_3^- arises from singly occupied $5P_z$ orbitals on two iodine atoms and a full $5P_z$ orbital on I^-, then there are three atomic orbitals involved. Thus, 3 molecular orbitals are formed, one bonding, one non-bonding and 1 antibonding. There are 4 e^-'s, 2 occupy bonding MO and 2 occupy the non-bonding MO. This gives one bond, spread over two positions i.e. the bond order of I–I bond in I_3^- is 0.5.

5. **(d)** $[Fe(CO)_5]$ is trigonal bipyramidal in shape. $[Ni(DMG)_2]$ is a rosy-red precipitate having square planar shape. $[Ni(CN)_4]^{2-}$ is colourless and diamagnetic with $\mu = 0$. $[Fe(CN)_5NO]^{2-}$ has strong field ligands allowing d^2sp^3 hybridization.

6. **(c)** $$R{-}OR' + H{-}I \longrightarrow R{-}OH + R'{-}I \xrightarrow[\text{solution}]{\text{Alcoholic } AgNO_3} AgI\downarrow + R'^{+}NO_3^-$$

Ether with one mole of methoxy group reacts with one mole of HI to form one mole of CH_3I, which can react with alcoholic $AgNO_3$ to give 1 mole of AgI precipitate. Similarly, 1 mole of ether with 'n' moles of methoxy group reacts with 'n' moles of HI to give 'n' moles of AgI precipitate.

Moles of organic compound = $\frac{0.2063}{168}$

Moles of AgI produced = $n \times \frac{0.2063}{168} = \frac{0.8658}{235}$

$n = 3.$

7. **(c)** SiC does not get hydrolysed as it is a covalent carbide. Na_2C_2 on hydrolysis gives C_2H_2 (So, it's an acetylide) and Mg_2C_3 on hydrolysis gives $CH_3C{\equiv}CH$ (So, it's a propynide). Mn_3C on hydrolysis liberates CH_4 and H_2, so it is a methanide.

$$Mn_3C + 6H_2O \longrightarrow 3Mn(OH)_2 + CH_4\uparrow + H_2\uparrow$$

8. **(d)** Let total number of moles of Ni are *x*, out of which 96% are in +2 state and 4% are in +3 state.

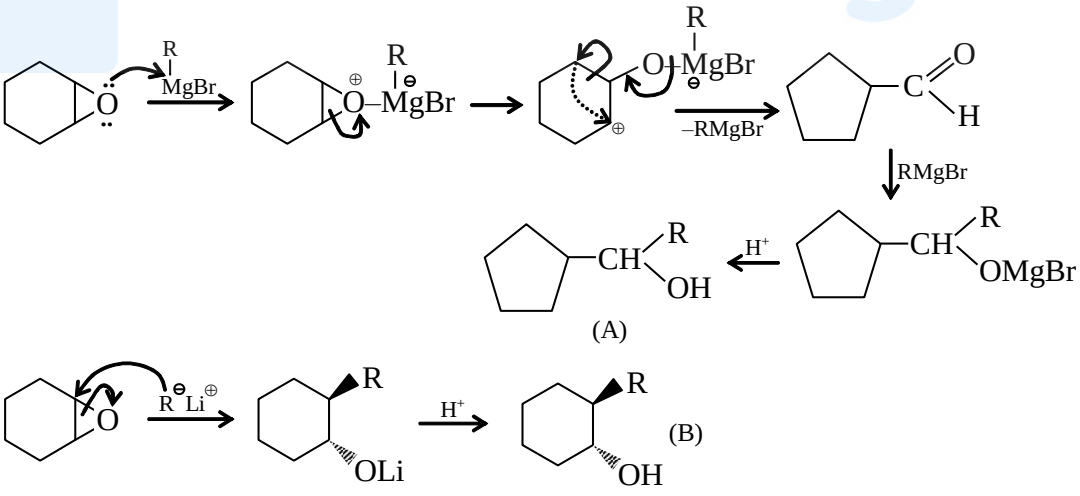
Maintaining the electroneutrality,

$$\left[x \times \frac{96}{100}(+2)\right] + \left[x \times \frac{4}{100}(+3)\right] = 2$$

$$x = \frac{200}{204} = \frac{50}{51} = \mathbf{0.98}$$

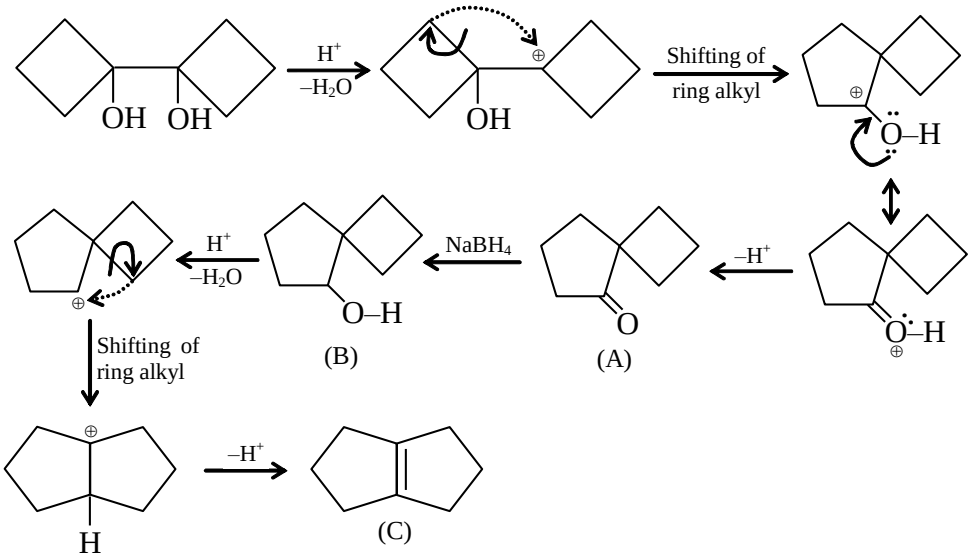
SECTION – II
Multiple Correct Choice Type

9. **(a, b)**



10. **(d)** ΔG at equilibrium is zero and thus cannot be related to K_p or K_c. It is $\Delta G°$, that is related to K_p or K_c, provided K_p and K_c both exist for a given equilibrium. For equilibrium involving solution species, K_p can not exist.

11. (b, c)



12. (a, b & d)

Let 'n' moles of A and B are present in solution. So, the moles of A in the solution is $\frac{n}{3}$.

Total number of moles in the distillate = $\frac{n}{4}$

Number of moles of A in the distillate = $\frac{n}{4}\times\frac{3}{4} = \mathbf{\frac{3n}{16}}$

Total pressure of solution, $P_T = P_A^o X_A + P_B^o X_B$

$$600 = 300 \times \frac{1}{3} + P_B^o \cdot \frac{2}{3}$$

$$\frac{3\times 500}{2} = P_B^o$$

$$P_B^o = \frac{1500}{2} = \mathbf{750\ mm\ Hg}.$$

Total number of moles in the residue = $\frac{3n}{4}$

Moles of A in the residue = $\frac{n}{3}-\frac{3n}{16} = \frac{7n}{48}$

Mole fraction of A in residue = $\frac{\frac{7n}{48}}{\frac{3n}{4}} = \frac{28}{48\times 3} = \frac{7}{36}$

Mole fraction of B in residue = $1 - \frac{7}{36} = \mathbf{\frac{29}{36}}$

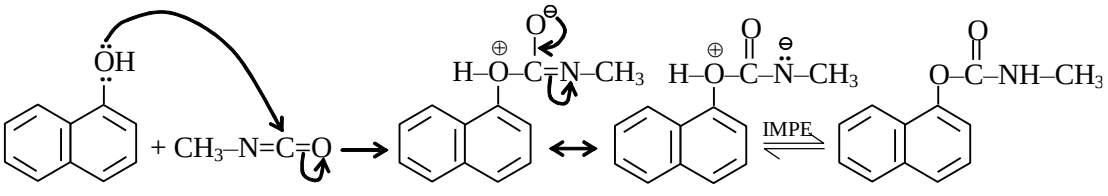
Total pressure of the residue $= \left(300\times\frac{7}{36}\right)+\left(\frac{1500}{2}\times\frac{29}{36}\right)$

$= 58.33 + 604.17 = \mathbf{662.5\ mm\ Hg}$.

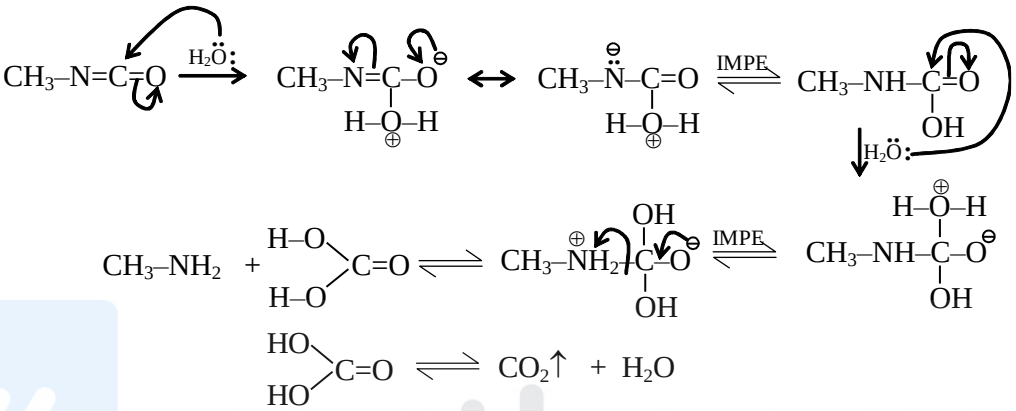
SECTION–III
Comprehension Type

Passage–I

13. **(b)**



14. **(c)**



15. **(d)** Due to intermolecular H-bonding between $CH_3–NH_2$ and H_2O, energy will be released, thus raising the temperature of the system.

Passage–II

16. **(b)** $f = \dfrac{99.9/0.1}{0.1/99.9} = (999)^2$

$$f' = \frac{r_{^{235}UF_6} / r_{^{238}UF_6}}{n_{A_o} / n_{B_o}} = \sqrt{\frac{M_{^{238}UF_6}}{M_{^{235}UF_6}}} = \sqrt{\frac{352}{349}}$$

$(f')^m = f$ where 'm' is the number of effusion steps needed to get desired concentration ratio.

$$\left(\frac{352}{349}\right)^{\frac{m}{2}} = (999)^2$$

$$\frac{m}{2}[\log 352 - \log 349] = 2 \log 999 = 2 \times 3$$

$$\frac{m}{2}[2.546 - 2.542] = 2 \times 3$$

$$m(0.004) = 12$$

$$m = \frac{12}{4\times10^{-3}} = 3 \times 10^3 = \textbf{3000 steps}.$$

17. **(c)** $f = \frac{64/1}{2/1} = 32 = 2^5$

$f' = \frac{r_{H_2}/r_{He}}{n_{H_2}/n_{He}} = \sqrt{\frac{M_{He}}{M_{H_2}}} = \sqrt{\frac{4}{2}} = \sqrt{2}$

$(f')^m = f$

$(2)^{m/2} = 2^5$

$\frac{m}{2} = 5$

$\therefore$ m = **10**

18. **(b)** $f = \frac{n_{H_2}/n_{D_2}}{10/90}$

$f' = \frac{r_{H_2}/r_{D_2}}{10/90} = \sqrt{\frac{M_{D_2}}{M_{H_2}}} = \sqrt{\frac{4}{2}} = \sqrt{2}$

$(f')^m = f$

$(2)^{m/2} = \frac{n_{H_2}/n_{D_2}}{10/90}$; $(2)^{18/2} \times \frac{10}{90} = \frac{n_{H_2}}{n_{D_2}}$

$2^9 \times \frac{1}{9} = \frac{n_{H_2}}{n_{D_2}}$; $\frac{512}{9} = 56.88 = \frac{n_{H_2}}{n_{D_2}}$

$\therefore$ % of H_2 in final mixture = $\frac{56.88}{57.88} \times 100 \simeq$ **98%**.

SECTION–IV
Matrix–Match Type

1. **(A) – 1, 2, 3, 4, 5 ; (B) – 3, 4, 5 ; (C) – 3, 4, 5 ; (D) – 2, 3**

2. **(A) – 1, 2, 3, 4, 5 ; (B) – 1, 2 ; (C) – 1, 2, 3, 4, 5 ; (D) – 2, 3, 5**